

Spectral Score Theoretical Layer v0.1 (Movie V0) — every SM particle organized through the dictionary by K-type Casimir and sector; substrate-primary anchors marked; quasi-eigentone framework spine; ready for Grace/Elie’s V0 artifact build.

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Spectral Score — theoretical layer for Movie V0

0. The artifact concept

Movie V0 (Casey’s framing) is the spectral score: every SM particle plotted at its K-type Casimir value, organized by sector, with the substrate-primary structure laid bare. This doc gives the theoretical layer — what each particle anchors to in the dictionary, what the spacings mean, where the decay arrows go.

1. The score layout

Vertical axis: SECTORS (sequenced by structural hierarchy): 1. SCALAR — $V_{(0,0)}$ trivial, Casimir 0 $\rightarrow$ Higgs 2. LEPTON-Shilov — $V_{(1/2,1/2)}$, Casimir $5/2 = \rho_1 \rightarrow$ leptons ($e, \mu, \tau, \nu \times 3$) 3. VECTOR-gauge — $V_{(1,0)}$, Casimir $4 = \text{rank}^2 \rightarrow$ photon, gluon 4. ADJOINT-gauge — $V_{(1,1)}$, Casimir $6 = C_2 \rightarrow$ W, Z, gluon-adjoint 5. BULK-fundamental — $V_{(1,1)} \text{bulk}_k=6$, Casimir $6 = C_2 \rightarrow$ proton, neutron, hadrons 6. Higher-Casimir composites — $V_{(2,0)}$, $V_{(3/2,1/2)}$, etc. $\rightarrow$ excited bulk states

Horizontal axis: Casimir value (or mass scale, logarithmic). Particles plotted at their K-type Casimir within each sector.

2. The substrate-primary anchors

The deep structure of the score: every “stable” particle’s K-type Casimir = a substrate primary.

K-type	Casimir	Substrate primary anchor	Particle(s)
$V_{(0,0)}$	0	trivial (vacuum)	Higgs (scalar)
$V_{(1/2,1/2)}$	$5/2 = \rho_1 = n_C/\text{rank}$	Bergman kernel singularity exponent	leptons (NATURAL MASS UNIT, L4)
$V_{(1,0)}$	$4 = \text{rank}^2$	canonical vector dim	photon ($V_{(1,0)}$ dim 5 = n_C)
$V_{(1,1)}$	$6 = C_2$	adjoint Casimir of B_2	gauge sector (W, Z, adjoint gluons)
$V_{(2,0)}$	$10 = \dim \text{so}(5)$	adjoint dim	(excited bulk)
$V_{(3/2,1/2)}$	$15/2$	$n_C/\text{rank} + N_c$	mixed-spinor states
$V_{(3/2,3/2)}$	$21/2 = \dim \text{so}(5,2)/2$	half-Lie-algebra-dim	composite spinor
$V_{(1,1)} \text{bulk } k=6$	$6 = C_2$	bulk fundamental (boundary $\rightarrow$ bulk bridge)	proton, neutron, baryons

The score's striking visual: the 4 SM fundamental sectors (Higgs scalar, lepton, photon, gauge) sit at Casimirs 0, 5/2, 4, 6 — all substrate primaries. Higher composites sit at Casimirs 10, 15/2, 21/2 — also substrate-natural products (dim so(5), n_C+5 , dim so(5,2)/2). The K-type Casimir spectrum IS the substrate's spectral lattice; SM particles populate specific lattice points.

3. The mass-scale spacings (within and across sectors)

The Casimir alone doesn't give masses; mass requires the kernel-integral structure. But the spacings within sectors give the mass-ratio structure:

Sector	Spacing relation	Source
Lepton radial (within $V_{(1/2,1/2)}$)	$m_\mu/m_e = (24/\pi^2)^6 \approx 206.77$	T190 (closed form via $24 = \text{rank}^3 \cdot N_c$, π^2 kernel power, $6 = C_2$)
Bulk-fundamental (within $V_{(1,1)}_{\text{bulk } k=6}$)	$m_\tau/m_\mu \approx 16.82$	T2003 (BST primary product) isospin splitting (small)
	$m_n - m_p \approx 1.3 \text{ MeV}$	
	$m_p/m_e = 6\pi^5 = C_2 \cdot \pi^{(n_C)}$	T187 (boundary→bulk bridge; 0.002% precision)
Gauge (EWSB)	$m_W/m_Z = \sqrt{(g/N_c)^2}$	Weinberg (derived)
PMNS lepton mixing	$\sin^2\theta_W = \text{rank}/N_c^2 = 2/9$	Weinberg (derived)
	$\sin^2\theta_{12} = \text{rank} \cdot N_c \cdot g / N_{\text{max}} = 42/137$	F1 falsifier (Elie Toy 3618)
	$\sin^2\theta_{23} = N_c \cdot n_C^2 / N_{\text{max}} = 75/137$	F1
	$\sin^2\theta_{13} = N_c / N_{\text{max}} = 3/137$	F1

The lepton-as-natural-unit alignment (lepton Casimir = ρ_1 = Bergman kernel exponent, L4) means the LEPTON sets the substrate's natural mass scale; all other masses are ratios from it via specific kernel-integral closed forms (T190, T2003, T187, etc.).

4. The decay arrows (quasi-eigentone framework spine)

The score has DOWNWARD arrows from QUASI-EIGENTONES (excited states) to their decay products (lower states). All arrows are Green-coproduct decompositions with grading conservation:

From	To	Conservation	Mechanism
μ	$e + \bar{\nu}_e + \nu_\mu$	Q, L preserved	$\Delta(\mu)$ coproduct
τ	many channels (hadronic + leptonic)	Q, L preserved	$\Delta(\tau)$ coproduct, multiple
n	$p + e + \bar{\nu}_e$	Q, B, L preserved	$\Delta(n) = \Delta(p) \otimes \text{leptonic (E3)}$
W^-	$\ell + \bar{\nu}$ or $q + \bar{q}'$	Q, generation preserved	$\Delta(W)$ gauge-fermion coproduct
Z	$f + \bar{f}$	Q, all charges 0	$\Delta(Z)$ gauge-fermion coproduct
H	$b + \bar{b}$, gluon-gluon, etc.	all charges 0	$\Delta(H)$ Yukawa/gauge coproduct
π^+	$\mu + \nu_\mu$	Q, L preserved	$\Delta(\pi)$ meson coproduct
K^0	$\pi + \dots$	Q preserved	$\Delta(K)$ meson coproduct
Λ	$p + \pi^-$	Q, B, strangeness	$\Delta(\Lambda)$ baryon coproduct

From	To	Conservation	Mechanism
(every hadron)	lower hadrons + leptons	grading preserved	$\Delta(\text{hadron})$ coproduct

All decays trace to Green-coproduct decompositions with the Hopf-algebra grading preserving Q, B, L . β -decay (E3) is the worked example; the framework generalizes to all SM decays.

5. The Five-Absence visualization

The spectral score should also show the GAPS — where the SM does NOT have particles, and why: - No 4th generation K-type slot — $V_{(1/2,1/2)}$ sector has no 4th radial level (closed at $h^{\nu-3}$ generations). - No SUSY K-type doublings — would require duplicating every K-type with shifted statistics (substrate doesn't predict this). - No GUT leptoquarks — would require a K-type connecting bulk-quark and Shilov-lepton sectors (substrate has Hardy-determinacy barrier). - No $0\nu\beta\beta$ K-type — would require ν Majorana K-type (substrate predicts Dirac per F5). - No magnetic monopole K-type — $U(1)$ is compact, no monopole topology.

These are visual ABSENCES in the score — predicted gaps that current observation confirms.

6. The score's narrative (for Movie V0)

The V0 movie can show:

1. Opening: the bare substrate D_{IV}^5 — empty Casimir spectrum.
2. K-type lattice appears: the spectral score's lattice points light up — every K-type at its Casimir.
3. SM particles populate: each SM particle drops onto its K-type slot — leptons at $V_{(1/2,1/2)}$, photon at $V_{(1,0)}$, W/Z at $V_{(1,1)}$, proton at bulk $k=6$, etc. Substrate-primary anchors label each.
4. Stable particles glow steady (true eigentones); quasi-eigentones pulse (decay arrows emanate to their products).
5. Decay arrows animate: $\mu \rightarrow e + \bar{\nu} + \nu$, $n \rightarrow p + e + \bar{\nu}$, $W \rightarrow \ell + \nu$, etc. — each arrow follows the Green coproduct with conservation labels.
6. Five-Absence highlight: predicted-empty slots remain dark — visualizing the substrate's predictions for what's NOT there.
7. Closing: the full SM as the substrate's spectral signature.

7. Honest scope + tier

RIGOROUS (substrate / engine theory): - K-type Casimirs (computed, anchored to substrate primaries). - Green coproduct decay mechanism (engine v0.1 + E3 + quasi-eigentone framework). - Substrate-primary alignments (L4 lepton anchor, T187 $6\pi^5$, Weinberg ratios, F1 PMNS).

FRAMEWORK (this doc): - Theoretical layer for V0 spectral score — particle placements, anchors, spacings, decay arrows, Five-Absence visualization. - Ready for V0 artifact build (Grace catalog + Elie computational + visual rendering).

OPEN (multi-week): - Quantitative decay rates from kernel matrix elements (L4 v0.2). - Per-particle mass derivations from radial-tower structure (L4 v0.2). - The bulk-color $SU(3)$ emergence for full quark sector placement.

Cal #27 / honesty: this v0.1 is the THEORETICAL layer for a visualization artifact — it doesn't derive new content but ORGANIZES the dictionary's existing placements + the quasi-eigentone framework into a score that can be visualized. The artifact build (Grace/Elie/visual) is the next step.

Routed: → Grace: V0 spectral score artifact build using this theoretical layer; Periodic Table v0.5 maps directly to the score. → Elie: when bulk K-type radial tower lands, the kernel matrix elements give the explicit mass spacings + decay rates. → Casey/team: V0 movie narrative spine is in place (substrate Casimir lattice + SM populates it + decay arrows + Five-Absence gaps). → me: continuing to Vol 16 chapter or EOD prep.

— Lyra, Spectral Score Theoretical Layer v0.1. Movie V0's structural backbone: every SM particle plotted at its K-type Casimir, organized by sector, anchored to substrate primaries (Casimir 0, 5/2, 4, 6, 10, ...), with quasi-eigentone framework providing decay arrows and Five-Absence framework providing predicted-empty gaps. Ready for V0 artifact build.